Reg. No.									

MANIPAL INSTITUTE OF TECHNOLOGY

MANIPAL

(A constituent unit of MAHE, Manipal)

VII SEMESTER B.TECH. (ELECTRICAL & ELECTRONICS ENGINEERING)

END SEMESTER EXAMINATIONS, NOVEMBER- DECEMBER 2023

DATABASE MANAGEMENT SYSTEMS [ELE 4079]

REVISED CREDIT SYSTEM

Time: 3 Hours **22 November 2024** **Max. Marks: 50**

Instructions to Candidates:

- Answer **ALL** the questions.
- Missing data may be suitably assumed.

1A. With the help of a suitable example differentiate between **DATA** and **INFORMATION**. ***03***

1B. What is **Single Tier client-server architecture**? Give a real-world example and explain. ***03***

1C. What is **Deadlock**? How is **deadlock** detected? Explain **deadlock** prevention schemes? ***04***

2A. A tropical fish hobbyist has six types of fish: Alphas (A), Betas (B), Certas (C), Deltas (D), Epsalas (E), and Fetas (F). Due to factors like predator-prey relationships, water conditions, and size, some fish cannot be kept together in the same tank. The hobbyist needs a database to track which fish are in each tank while minimizing the number of tanks used.

The following table shows which fish cannot be together:

Type	A	B	C	D	E	F
Cannot be with	B, C	A, C, E	A, B, D, E	C, F	B, C, F	D, E

Give a suitable **Entity-Relationship (ER)** model using proper notations for creating the database. ***03***

2B. A tropical fish hobbyist has six types of fish: Alphas (A), Betas (B), Certas (C), Deltas (D), Epsalas (E), and Fetas (F). Due to factors like predator-prey relationships, water conditions, and size, some fish cannot be kept together in the same tank. The hobbyist needs a database to track which fish are in each tank while minimizing the number of tanks used.

The following table shows which fish cannot be together:

Type	A	B	C	D	E	F
Cannot be with	B, C	A, C, E	A, B, D, E	C, F	B, C, F	D, E

Some fish are **predators (**eating other fish**),** and others are **passive** (Herbivorous, eating plants), using the concepts of **SPECIALISATION** and **GENERALISATION** give a suitable **Extended Entity-Relationship (ER) model** using proper notations for creating the database with additional features. ***03***

2C. A tropical fish hobbyist has six types of fish: Alphas (A), Betas (B), Certas (C), Deltas (D), Epsalas (E), and Fetas (F). Due to predator-prey relationships, water conditions, and size, some fish can be kept together. Some fish are predators, while others are passive, ***04***

herbivorous types that only eat plants. The hobbyist needs a database to track which fish are in each tank while minimizing the number of tanks used.

The following table shows which fish cannot be together:

Type	A	B	C	D	E	F
Cannot be with	B, C	A, C, E	A, B, D, E	C, F	B, C, F	D, E

Transform the Extended Entity Relationship model developed in **Q2B** for the above requirement to its equivalent **Relational Model/Schema**. With proper notations, give the relational schema for the entities and relationships identified.

3A. Consider the following table (relation), by applying concepts of relational database design convert it to 1NF, 2NF and 3NF (Normal Forms).

Customer Name	**Purchased Item**	**Date of Purchase**
John Doe	Baked Beans	10-Jan-24
John Doe	Bread	12-Feb-23
Jeff Smith	Milk	29-May-20
Jeff Smith	Baked Beans	12-Dec-23
Stephen Jones	Bread	17-Apr-22
Stephen Jones	Milk	22-Aug-24

03

3B. Consider the following schema:

- Suppliers (<u>supplierid</u>: **integer**, sname: **string**, address: **string,** city**: string**)
- Parts (<u>partid</u>: **integer**, productname: **string**, color: **string**)
- Catalog (<u>supplierid</u>: **integer**, <u>partid</u>: **integer**, cost: **decimal**)

The key fields are underlined, and the domain of each field is listed after the field name. Thus, supplierid is the key for Suppliers, partid is the key for Parts, and supplierid and partid together form the key for Catalog. The Catalog relation lists the prices charged for parts by Suppliers.

Write **RELATIONAL ALGEBRA** queries for the following:

a) Find the product names of all red coloured parts.
b) List all suppliers who belong to New Delhi city.
c) Illustrate the use of intersection operation for finding product names of red and green coloured parts.

03

3C. Consider the following schema:

- Suppliers (<u>supplierid</u>: **integer**, sname: **string**, address: **string,** city**: string**)
- Parts (<u>partid</u>: **integer**, productname: **string**, color: **string**)
- Catalog (<u>supplierid</u>: **integer**, <u>partid</u>: **integer**, cost: **decimal**)

The key fields are underlined, and the domain of each field is listed after the field name. Thus, supplierid is the key for Suppliers, partid is the key for Parts, and supplierid and partid together form the key for Catalog. The Catalog relation lists the prices charged for parts by Suppliers.

Write **RELATIONAL ALGEBRA** queries for the following:

a) Find the total number of suppliers
b) Find the maximum cost of any part supplied in the Catalog.
c) Get the names and colors of parts supplied by supplier with ID 200.
d) Get the part names and supplier names for parts with a cost greater than 100.

04

4A. A sample create table SQL statement is given below for reference: ***03***

```
CREATE TABLE Orders (
   OrderID int NOT NULL,
   OrderNumber int NOT NULL,
   PersonID int,
   PRIMARY KEY (OrderID),
   FOREIGN KEY (PersonID) REFERENCES Persons(PersonID)
);
```

Based on the sample query given, write **SQL statements** to create tables for following schemas:

- Suppliers (supplierid: **integer**, sname: **string**, address: **string,** city **: string**)
- Parts (partid: **integer**, productname: **string**, color: **string**)
- Catalog (supplierid: **integer**, partid: **integer**, cost: **decimal**)

Assume data types as per requirement.

The key fields are underlined, and the domain of each field is listed after the field name. Thus supplierid is the key for Suppliers, partid is the key for Parts, and supplierid and partid together form the key for Catalog. The Catalog relation lists the prices charged for parts by Suppliers.

4B. Consider the following relation **Employee**:

Employee

EmpID	EmployeeName	EmpSalary
1	Jannik Sinner	1500
2	Carlos Alcaraz	1200
3	Novak Djokovic	1000

The relation has the following structure
Employee
EmpID tinyint(4) PRIMARY KEY auto_increment NOT NULL
EmployeeName varchar (50) NOT NULL
EmpSalary int (11) NOT NULL

Sample statements for reference:

- INSERT INTO *table_name* (*column1*, *column2*, *column3*, ...) VALUES (*value1*, *value2*, *value3*, ...);
- UPDATE *table_name* SET *column1 = value1*, *column2 = value2*, ... WHERE *condition*;
- DELETE FROM *table_name* WHERE *condition*;

Write **SQL statements** for the following, these statements will be executed in the same sequence:

1) Insert new employees 'Roger Federer' with a salary of 2000 and 'Rafael Nadal' with a salary of 1800 into employee table using **SINGLE SQL STATEMENT**.
2) Increase salaries by 10% of those employees whose salary is less than 1500. How many employees are affected by this increment.
3) Delete employee whose Name begins with letter 'R'. **03**

4C. Consider the following relation **Employee**:

Employee

EmpID	EmployeeName	EmpSalary
1	Jannik Sinner	1500
2	Carlos Alcaraz	1200
3	Novak Djokovic	1000

The relation has the following structure
Employee **04**

EmpID tinyint(4) PRIMARY KEY auto_increment NOT NULL
EmployeeName varchar(50) NOT NULL
EmpSalary int(11) NOT NULL

Sample select statement for reference:
SELECT *column_name(s)* FROM *table_name* WHERE *condition*
GROUP BY *column_name(s)* ORDER BY *column_name(s);*

Write **SQL statements** for the following:

1) Retrieve the names of employees who have a salary greater than 1200 order the records in the descending order of salary.
2) What is the highest salary among all employees?
3) Select the names of employees whose names start with the letter "N".
4) Select the count of employees earning more than 1000.

5A. What are the **hardware** and **software** requirements to implement a temperature monitoring system that records temperature every minute and counts occurrences of temperatures above 25°C daily, with the ability to store, process, and report this data effectively? ***03***

5B. For implementing a temperature monitoring system that records temperature every minute and counts occurrences of temperatures above 25°C daily, give the conceptual, logical and physical model designs. ***03***

5C. For implementing a temperature monitoring system that records temperature every minute and counts occurrences of temperatures above 25°C daily, give the Entity-Relationship and relational model/schema as per proper notations. ***04***